

TAYYAB BAHADUR

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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on experience in React.js, Node.js, and ASP.NET, plus applied experience in Machine Learning and AI-driven systems including RAG pipelines, computer vision, and signal processing. Proven track record delivering real-world projects such as an EEG-based sleep monitoring system, drone detection using OpenCV, and local AI agents. Strong problem-solving skills with experience in SQL optimization, backend performance, and end-to-end application development.

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, C++, Assembly
Frontend: React.js, Next.js, HTML, CSS, Bootstrap
Backend & DB: Node.js, ASP.NET, MySQL, Firebase
Tools & Concepts: GA4, GTM, Android Studio, Visual Studio, IntelliJ, Figma, OOP, Data Structures

EDUCATION

BS Computer Engineering | FAST University, Lahore **Aug 2022 – June 2026**
Relevant Coursework: Data Structures & Algorithms, Operating Systems, OOP, Applied Machine Learning, Database Systems, Embedded Systems, Cloud Computing.

Intermediate (Pre-Engineering) | Govt. College Civil Lines, Lahore **Jan 2020 – May 2022**

EXPERIENCE

Software Engineering Intern | LOXVO Technologies **June – Aug 2023**

- Designed and developed scalable full-stack applications utilizing React.js, Node.js, and JavaScript.
- Created user-friendly, responsive front-end layouts using Bootstrap and CSS.
- Managed SQL databases, writing efficient queries for data storage and optimizing backend performance.

Web Developer & Google Analytics Intern | INVOLVZ **June – Aug 2025**

- Built responsive, cross-browser compatible web interfaces using React.js, HTML, CSS, and Bootstrap.
- Managed and customized WordPress websites to meet specific client design requirements.
- Integrated Google Analytics 4 (GA4) and GTM to track user engagement and optimize digital reach.

Vice President | FAST Sports Society **2025 – 2026**

- Organized inter-university sports events and tournaments for student participation.
- Planned and managed corporate sporting events in collaboration with partner companies.
- Led management and operations teams, overseeing logistics, scheduling, and on-ground execution for events.

KEY PROJECTS

SleepTrack – Smart Non-Invasive Sleep Monitoring System (Final Year Project)

Stack: React.js, Node.js, Python, AWS, MongoDB, Machine Learning

- Developed an EEG-based sleep monitoring system with automated cloud processing and sleep-stage classification.
- Built a React.js dashboard and Node.js backend for session management and result visualization.
- Implemented machine learning models and signal-processing pipelines for EEG feature extraction and analysis.
- Integrated AWS services, MongoDB, and embedded hardware to create a complete end-to-end solution.

Local AI Document Assistant (RAG-based)

Stack: Python, Ollama, ChromaDB, SentenceTransformers

- Built a local AI agent using Retrieval-Augmented Generation (RAG) with Ollama for offline LLM inference.
- Implemented document ingestion pipeline supporting multiple file formats with batch embedding generation.
- Integrated ChromaDB for vector storage and retrieval with cached embedding models for improved performance.

Drone Detection System

Stack: Python, OpenCV, Computer Vision

- Designed and developed a real-time drone detection system using Python and OpenCV.
- Implemented object detection, tracking, and image processing techniques to identify aerial objects from video feeds.
- Processed and analyzed video streams for drone monitoring and visualization.
- Enhanced detection performance through image preprocessing and feature extraction methods.

Smart Home Management System

Stack: ASP.NET, MySQL, Bootstrap, HTML, CSS, JS

- Developed a web-based management system for smart home devices using .NET framework.
- Designed a relational database schema in MySQL to handle user data and device states efficiently.
- Implemented a responsive dashboard for real-time monitoring and control.

React Web Application Suite

Projects: TODOApp, TextUtils, Expense Tracker (React, HTML, CSS, Bootstrap, JavaScript, MySQL)

- Expense Tracker:** Dynamic app to track income and expenses with visual data representation.
- TextUtils:** Utility app for text manipulation (analysis, conversion, and formatting).
- TODO App:** Task management interface with Create, Read, Update, and Delete (CRUD) functionality.